3. Input parameters implied by the term-by-term solution vs. Table 1 of universe_bestfit_thinshell.pdf. Each column keeps that fit's shell assumption. Newtonian Ω_s free: $\Omega_s = 1.012$, thin shell $R_{\rm in} = D - R$, $R_{\rm out} = D + R$. Newtonian Ω_s=0.049: $R_{\rm in} = D - 1.028\,R$, $R_{\rm out} = D + 1.475\,R$. The unknowns are solved from the Sec. 2 equations (2ℓ phase $\to \ell_0$, 2ℓ amplitude $\to \beta \to M_c/M_{\rm crit}$, $1\ell \to$ Amp, φ, 3ℓ cos $\to \varepsilon Q \to D$; C_m, S_m, m = 1–3, by least squares). The $a_m(\beta,\gamma)$ come from the code's exact $h(q)$, not the linear formula. Masses use $H_0 = 70$, $\rho_s = \Omega_s \rho_c$.

	Newtonian, Ω_s free		Newtonian, Ω_s=0.049	
Quantity	term-by-term	Table 1	term-by-term	Table 1
D/R	6.474	6.554	2.589	3.439
D	2.857×10^{24} km (92.6 Gpc)	2.892×10^{24} km (93.7 Gpc)	1.143×10^{24} km (37.0 Gpc)	1.517×10^{24} km (49.2 Gpc)
$R_{\rm in}$	78.3 Gpc	79.4 Gpc	22.3 Gpc	34.5 Gpc
$R_{\rm out}$	106.9 Gpc	108.0 Gpc	58.1 Gpc	70.3 Gpc
M_c	2.38×10^{54} kg $1.20\times10^{24}\ M_\odot$	2.47×10^{54} kg $1.24\times10^{24}\ M_\odot$	9.41×10^{52} kg $4.73\times10^{22}\ M_\odot$	2.31×10^{53} kg $1.16\times10^{23}\ M_\odot$
$M_c/M_{\rm crit}(D)$	0.00265	0.00265	0.00164	0.00171
$M(<D)$, total	3.62×10^{56} kg	3.72×10^{56} kg	2.29×10^{54} kg	4.56×10^{54} kg
$M_{\rm shell}$, total	8.50×10^{56} kg	8.71×10^{56} kg	1.03×10^{55} kg	1.70×10^{55} kg
M within R	3.35×10^{54} kg	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg
Ω_s	1.012[a]	1.012	0.049[a]	0.049
ℓ_0	13.2°	12.8°	12.8°	13.0°
Amp	−0.1246	−0.1240	−0.0637	−0.0622
φ	1.267 rad	1.264 rad	0.941 rad	0.929 rad
R^2 vs Migkas	0.909	0.909	0.881	0.893
R^2 vs A_F	0.987	0.987	0.970	0.977

[a] Held fixed (that column's shell assumption). The harmonic solve weights C_m, S_m equally, not the 71 points, so its R^2 can sit slightly below the point fit.

4. Model vs. generally accepted values. Model columns use the Table 1 values of the two Newtonian fits (their saved results and code; $H_0 = 70$). $R = 14.3$ Gpc is the comoving radius of the observable universe.

Quantity	Accepted value	Source	Newt. Ω_s free (0.909)	Newt. Ω_s=0.049 (0.893)
Ordinary-matter mass within R	1.51×10^{53} kg	[1][a]	3.35×10^{54} kg (22×)	1.62×10^{53} kg (1.07×)
Total matter mass within R	9.67×10^{53} kg	[1][a]	3.35×10^{54} kg[b]	1.62×10^{53} kg[b]
Ordinary-matter density ρ_b	4.20×10^{-28} kg m^{-3}	[1]	9.32×10^{-27} kg m^{-3}	4.51×10^{-28} kg m^{-3}
Ω_b	0.0493	[1]	1.012 (= Ω_s)	0.049 (= Ω_s)
Total matter Ω_m	0.3153 ± 0.0073	[1]	1.012 (= Ω_s, no dark matter)	0.049 (= Ω_s, no dark matter)
Critical density ρ_c	8.52×10^{-27} kg m^{-3} at H_0=67.36	[1]	9.20×10^{-27} kg m^{-3} at H_0=70	9.20×10^{-27} kg m^{-3} at H_0=70
H_0 [km s^{-1} Mpc^{-1}]	67.36 ± 0.54 (CMB); 73.04 ± 1.04 (local)	[1]; [2]	70 (input)	70 (input)
Earth (Sun) speed relative to the CMB	369.82 ± 0.11 km s^{-1}	[3]	$V_O = kD = 5.90\times10^6$ km s^{-1} = 19.7 c	$V_O = kD = 3.24\times10^6$ km s^{-1} = 10.8 c
ratio V_O/(CMB dipole speed)	1		15,958	8,757
Local Group speed relative to the CMB	620 ± 15 km s^{-1}	[3]	– (not modelled)	– (not modelled)
Bulk flow within 150 h^{-1} Mpc	395 ± 29 km s^{-1}	[4]	5.7 km s^{-1} c[c]	0.3 km s^{-1} c[c]
Core mass M_c (vs. ordinary mass within R)	–		2.47×10^{54} kg (16×)	2.31×10^{53} kg (1.5×)
Mass inside D (vs. ordinary mass within R)	–		3.72×10^{56} kg (2,460×)	4.56×10^{54} kg (30×)
Age / k^{-1}	13.797 ± 0.023 Gyr	[1]	k^{-1} = 13.97 Gyr; integration age 13.83 Gyr	k^{-1} = 13.97 Gyr; integration age 13.89 Gyr

[a] Derived from [1]: $\Omega_b h^2 = 0.02237$, $\Omega_m = 0.3153$, $H_0 = 67.36$, times the volume of the $R = 14.3$ Gpc sphere. Ratios in brackets: model / accepted ordinary-matter mass within R. [b] The model has no dark matter, so total = ordinary. [c] Mean of $\Delta\vec{V}$ over a 150 h^{-1} Mpc ball around Earth ($h = 0.70$), from the code's exact expansion-rate profile $h(q)$ (second order in radius/D; the first-order term averages to zero). The sine factor is not a velocity and is not included.

Sources. [1] Planck Collaboration 2020, A&A, 641, A6 (Planck 2018 VI, TT,TE,EE+lowE+lensing). [2] Riess, A. G., et al. 2022, ApJL, 934, L7 (SH0ES). [3] Planck Collaboration 2020, A&A, 641, A1 (Planck 2018 I; Local Group value after Kogut et al. 1993, ApJ, 419, 1). [4] Watkins, R., et al. 2023, MNRAS, 524, 1885 (CosmicFlows-4). Sedov–Taylor: Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159; Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (Academic Press).